

LUMEN

Team: Creators

Ashish Beniwal - ashish.beniwal@students.iiit.ac.in
IndieConnect ID: ashishbeniwal

Ishit Agarwal - ishit.agarwal@students.iiit.ac.in
IndieConnect ID: hyperterror

Princy Patel - princy.patel@students.iiit.ac.in
IndieConnect ID: tinchu

Moiesha Gupta - moiesha.gupta@students.iiit.ac.in
IndieConnect ID: ascian

Game Concept

Game Genre: Dark Fantasy Action Platformer

Core Gameplay Loop: The player controls Lumen, a mysterious light-bearing character exploring a corrupted world. Light is used as both a gameplay mechanic and a visual storytelling device. The player fights corrupted creatures, navigates ruined environments, and gradually restores color and light to the world by finding elemental Cores.

Theme Integration

Light: Light is the central gameplay mechanic. Lumen carries a light source and utilizes light energy during exploration, dynamic combat against corrupted enemies, and the environmental restoration of the dark world.

Comic: The game features a comic-book presentation layer overlaying its dark-fantasy art direction. While the core assets are pixel-art, the presentation utilizes comic-panel inspired framing, dramatic action cuts, impact flashes, stylized attack effects, and narrative panels to highlight key action moments.

Twist: The narrative twist centers around Lumen's relationship with the corrupted world and the light. The player initially believes they are simply restoring a dead world, but later discovers that the source of the corruption and Lumen's own purpose are intimately connected.

Gameplay & Visual Direction

Lumen is a 2D side-scrolling platformer requiring precision movement and timing. Combat involves striking enemies to build up a dynamic 'LightPower' meter, which is then expended for special attacks (e.g., throwing Ice Projectiles). The visual direction combines a dark-fantasy pixel art aesthetic (ancient ruins, dark stone platforms, glowing cyan/blue/green crystals) with a comic-inspired presentation layer for dramatic emphasis.

Technical Implementation

Current Prototype Status:

The current working prototype is built using Godot 4 (GDScript). It is a fully functional 2D action platformer with integrated pixel-art assets, complex collision, state machines, and a dynamic light/combat system. This serves as the definitive mechanical and visual design prototype.

Intended Web-Game Architecture (Future Implementation):

The final target architecture for this game is a native JavaScript Web Game to fulfill the technical requirements. The following technologies will be utilized in the final build:

1. Three.js:

Will be used for rendering the 2D/2.5D visual layers in WebGL, applying custom shaders for lighting, and rendering the comic-book style presentation layers (panels, flashes) dynamically over the game world.

2. InstancedBufferGeometry:

Will be utilized to render massive amounts of environmental particles (spores, snow, dust) and large repeating tilemaps efficiently in WebGL without creating excessive draw calls.

3. JavaScript Physics (Matter.js):

Matter.js will replace Godot's physics engine. It will handle the 2D character controllers, rigid body collisions, and platforming logic natively in JavaScript, ensuring smooth physics execution in the browser.

Asset Pipeline:

The generated visual assets (transparent sprite sheets, environment tiles) and conceptual gameplay logic (state machines, light mechanics) developed in the Godot prototype will be directly ported and reused in the Three.js/Matter.js architecture.